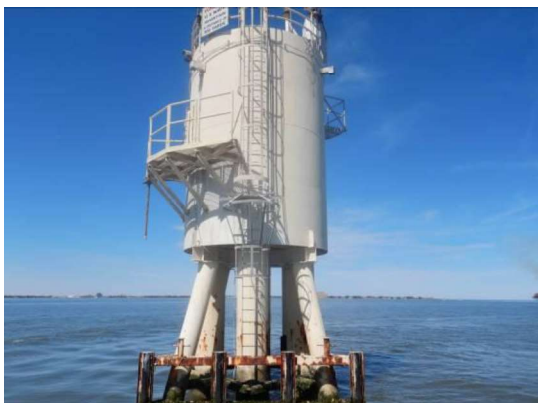


Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

Naval Station Norfolk
Norfolk, VA**Section 3.28 - Q58 Data Concentrator Shed****Contract:** N39430-15-D-1623**Inspection Date:** 03/05/2019**Contractor:** Collins Engineers, Inc.
123 N. Wacker Drive, Suite 900
Chicago, IL 60606**Facility:** Q58 Data Concentrator Shed**INFADS:** NFA100001163536**PRN:** 201157**Location:** [36.97745, -76.315354](#)**Estimated Repair Cost*:** \$ 129,920*Repair cost includes Design Allowances, Contractor Overhead & Profit, and Inflation Allowances.
See Appendix C - Cost Estimate for detailed analysis.

	Condition Index (CI)	75	Max Water Current	0.5kn
	Engineering Assessment Rating	Satisfactory	Water Clarity	3ft
	Operational Rating	C1	Tide Variation	4ft
	5 Year Projected CI	70	Max Water Depth	9ft
	10 Year Projected CI	65	Seasonal Water Temp	75°F
	Year(s) Previously Inspected	2013	Seasonal Ambient Temp	88°F

Facility Usage Description:

The Q58 Data Concentrator Shed is a steel structure used during degaussing range operations. The structure is supported on four steel pipe piles. There are two small bollards and a fender system for small craft berthing.

Summary of Repair Recommendations:

It is recommended that the steel assets above MLW be cleaned and recoated at this time.

Impact To Mission If Repairs Not Provided:

Significant impacts to the facility mission are not anticipated if repairs are not performed prior to the next scheduled routine inspection. However, the protective coating is failed along the above water portions of the steel assets and the noted corrosion will continue to progress and eventually lead to loss of steel section.

Operational Restrictions:

There are currently no operational restrictions to the Q58 Data Concentrator Shed.

Additional Facility Photos



Photo 3.28-1: Overall view of Q58 Data Concentrator Shed, looking north.



Photo 3.28-2: Overall view of Q58 Data Concentrator Shed, looking south.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.



Photo 3.28-3: Overall view of Q58 Data Concentrator Shed, looking northeast.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

H1010 - Substructure		Repair Cost \$42,770	CI: 75
H101001 - Pile Foundations		Repair Cost \$22,750	CI: 75
UFII Component	H10100101	Piles Asset Type(s): PB	
Findings	The steel piles are in Satisfactory Condition. There are four battered 18 in. steel piles. The steel piles have minor coating failure on up to 10% of the surface area and light surface corrosion in the tidal zone. Ultrasonic thickness measurement readings taken at the piles range between 0.480 in. and 0.495 in., and the potential reading is -0.680 volts. The anode sled was not located and is believed to be buried under silt and/or the cable bundles below the shed. There are no other notable defects.		
Recommendations	It is recommended that the steel piles be cleaned and recoated above MLW. It is also recommended that further investigations should be conducted to determine the location and condition of the anode sled.		

Supporting Photos

Photo Type: Typical UFII/Asset Type: H10100101_PB



Photo 3.28-4: View of the typical condition of the steel batter piles.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

H1010 - Substructure		Repair Cost \$42,770	CI: 75
H101002 - Pile Caps		Repair Cost \$20,020	CI: 75
UFII Component	H101002	Pile Caps Asset Type(s): PC1	
Findings	The steel pile caps are in Satisfactory Condition. There are two steel pile caps that are 15 ft long by 6 in wide. The pile caps have minor coating failure on up to 10% of the surface with light surface corrosion. There are no other notable defects.		
Recommendations	It is recommended that the steel pile caps be cleaned and recoated.		

Supporting Photos

Photo Type: Typical UFII/Asset Type: H101002_PC1



Photo 3.28-5: View of the typical condition of the steel pile caps.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

H1020 - Superstructure		Repair Cost \$62,790	CI: 75
H102001 - Beams and Girders		Repair Cost \$26,390	CI: 75
UFII Component	H102001	Beams and Girders Asset Type(s): B1, B2, & B3	
Findings	The steel beams are in Satisfactory Condition. There are eight steel beams. The beams and girders have minor coating failure on up to 10% of the surface area with light surface corrosion. There are no other notable defects.		
Recommendations	It is recommended that the steel beams be cleaned and recoated.		

Supporting Photos

Photo Type: Typical UFII/Asset Type: H102001_B1



Photo 3.28-6: View of the typical condition of the lower steel beams.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk
CR-NAVFAC EXWC-CIOFP - DRAFT Collins Engineers, Inc.

Photo Type: Typical UFII/Asset Type: H102001_B1



Photo 3.28-7: View of the typical condition of the floor beams.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

H1020 - Superstructure		Repair Cost \$62,790	CI: 75
H102002 - Columns		Repair Cost \$36,400	CI: 75
UFII Component	H102002	Columns Asset Type(s): COL	
Findings	The steel pipe columns are in Satisfactory Condition. There are four 30 in. steel columns. The columns have minor coating failure on up to 10% of the surface area with light surface corrosion. There are no other notable defects.		
Recommendations	It is recommended that the steel pipe columns be cleaned and recoated above MLW.		

Supporting Photos

Photo Type: Typical UFII/Asset Type: H102002_COL

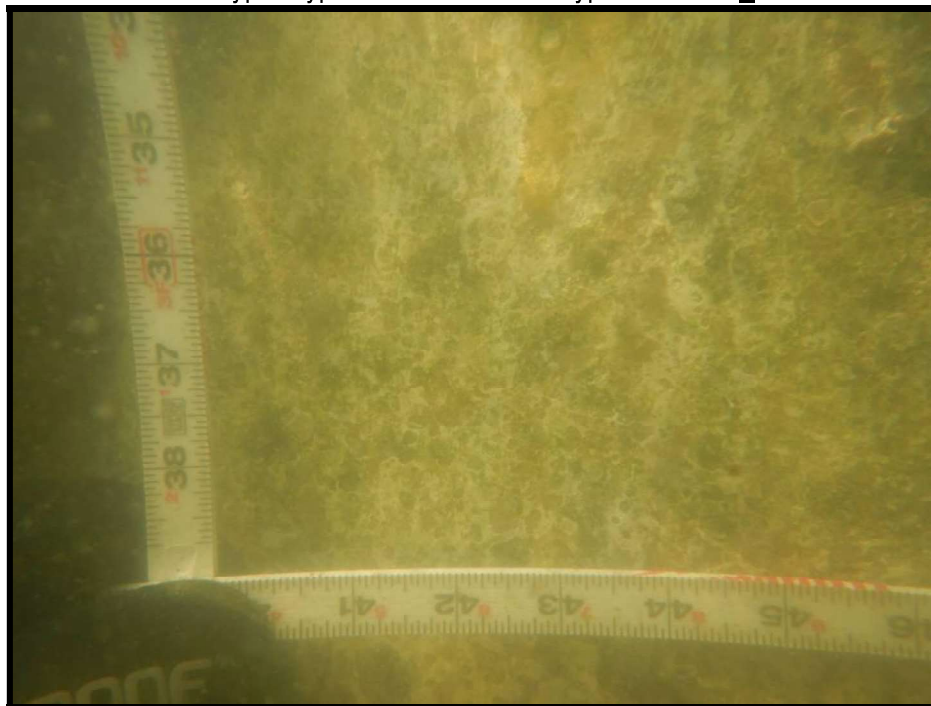


Photo 3.28-8: View of the typical condition of the steel columns below water after a Level II cleaning.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

Photo Type: Typical

UFII/Asset Type: H102002_COL



Photo 3.28-9: View of the typical condition of the steel pipe columns used for sensor cables.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

H1040 - Mooring & Berthing Systems**Repair Cost \$23,450****CI: 60****H104001 - Primary Fender System****Repair Cost \$21,630****CI: 60**

UFII Component	H10400102	Fender Framing Asset Type(s): FF1, FF2, & FF3
Findings	The fender frame members are in Fair Condition. There are two horizontal steel beams and four vertical steel channels. Each of the vertical steel channels has a rubber rubrail attached. The vertical rubber rubrails have minor abrasion damage. The steel members have moderate to major coating loss with moderate corrosion and delaminations. The rubrail mounting channels have moderate to heavy surface corrosion, with 100% section loss on the steel connection plates for the rubrails.	
Recommendations	It is recommended that the steel framing members be cleaned and recoated and that the steel connection plates be replaced.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H10400102_FF1

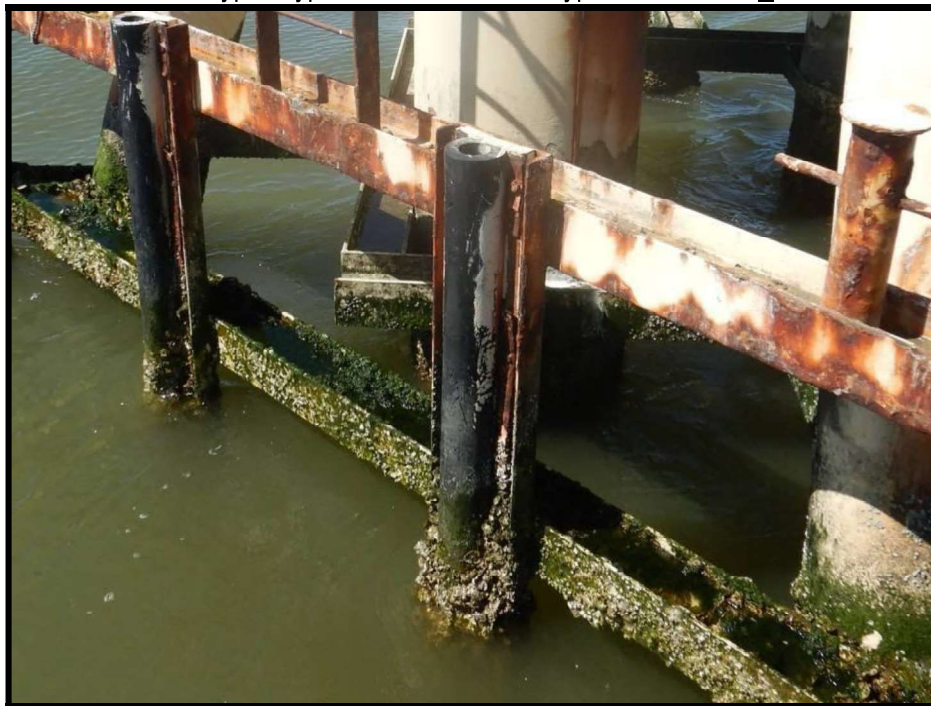


Photo 3.28-10: View of the typical coating failure and areas of moderate corrosion on the horizontal and vertical steel members and the minor abrasion damage along the rubber rubrails.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

Photo Type: Typical

UFII/Asset Type: H10400102_FF3



Photo 3.28-11: View the typical coating failure and moderate corrosion on the vertical steel channels and steel connection washers for the rubrails.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT Collins Engineers, Inc.

H1040 - Mooring & Berthing Systems **Repair Cost \$23,450** **CI: 60**

H104005 - Mooring Hardware **Repair Cost \$1,820** **CI: 60**

UFII Component	H10400501	Bollards Asset Type(s): MF
Findings	The steel bollards are in Fair Condition. There are two bollards that have up to 75% coating failure with moderate surface corrosion.	
Recommendations	It is recommended that the steel bollards be cleaned and recoated.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H10400501_MF



Photo 3.28-12: View of the typical coating failure and areas of moderate corrosion on the steel bollards.

Waterfront Facilities Inspections & Assessments at Naval Station Norfolk

CR-NAVFAC EXWC-CIOFP - DRAFT

Collins Engineers, Inc.

H1050 - Appurtenances		Repair Cost \$910	CI: 60
H105005 - Safety Ladders		Repair Cost \$910	CI: 60
UFII Component	H105005	Safety Ladders Asset Type(s): LAD	
Findings	The safety ladder is in Fair Condition. The ladder has coating failure and moderate surface corrosion and delamination on the bottom three rungs and rails.		
Recommendations	It is recommended that the safety ladder be cleaned and recoated.		

Supporting Photos

Photo Type: Typical UFII/Asset Type: H105005_LAD

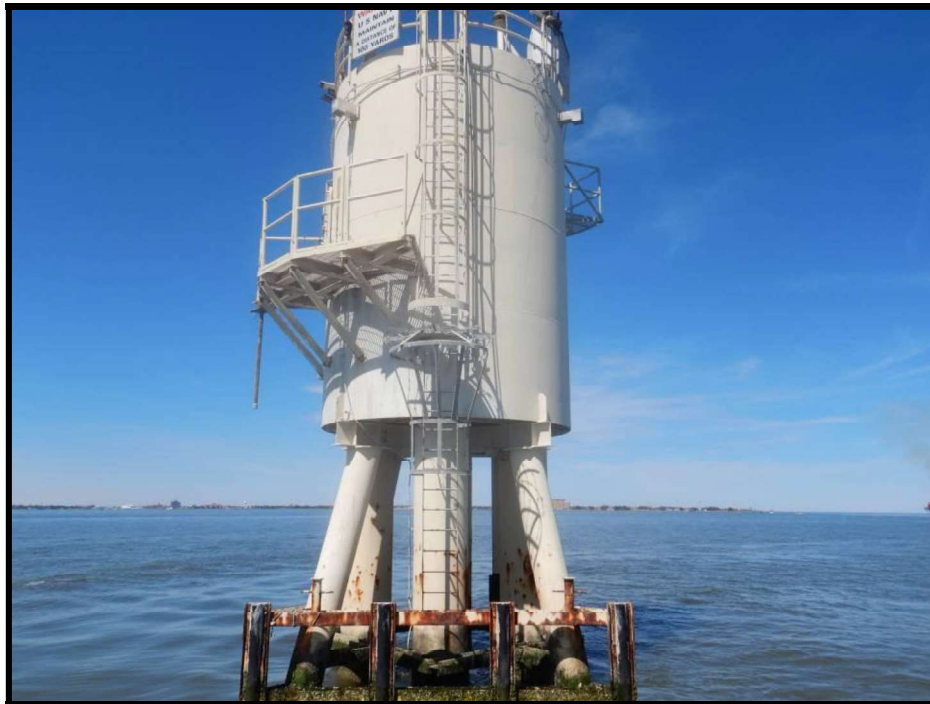


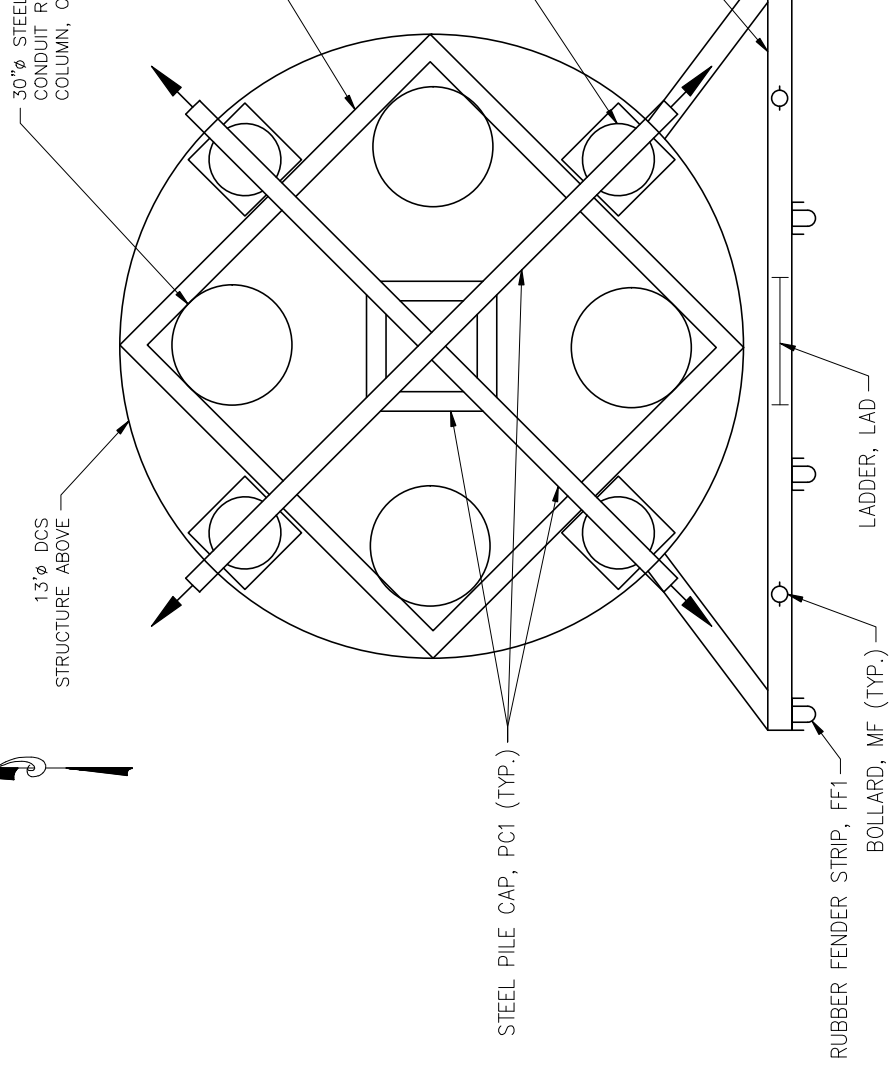
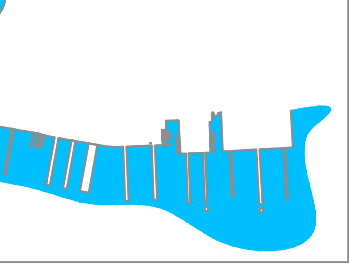
Photo 3.28-13: View of the safety ladder condition.

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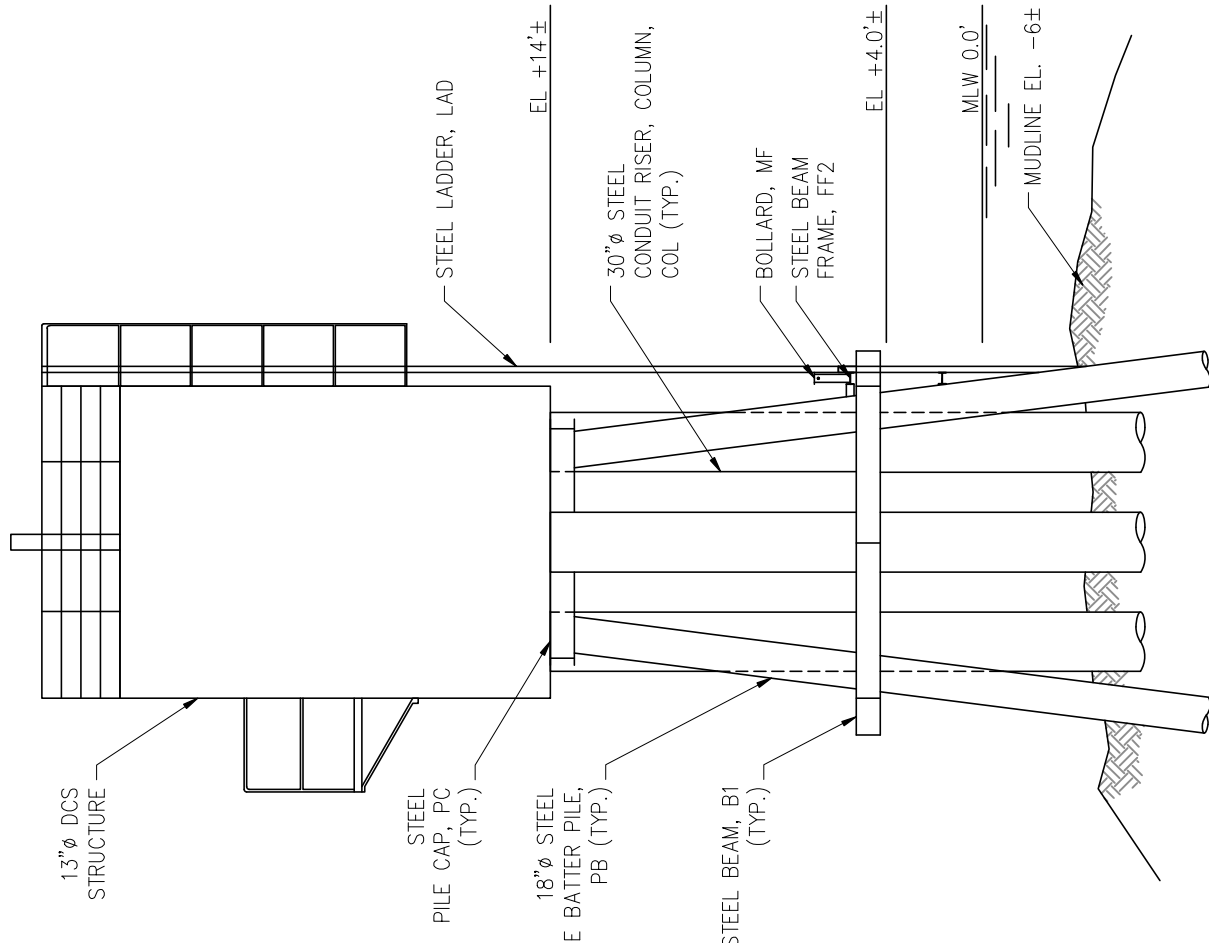
Photo Type: Defect UFII/Asset Type: H105005_LAD



Photo 3.28-14: View of the coating failure and corrosion on the bottom of the safety ladder.



PLAN
SCALE: $\frac{1}{4}" = 1'$



ELEVATION
SCALE: $\frac{1}{8}" = 1'$

NOTE: STEEL FLOOR BEAMS, B2 B3 NOT SHOWN FOR CLARITY